RESEARCH ARTICLE

Long-Term Outcomes After Ischemic Stroke Across Atrial Fibrillation Phenotypes Defined by Detection Timing and Prior Anticoagulation Status

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Neurology® 2026;107:e218198. doi:10.1212/WNL.0000000000218198

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Abstract

Background and Objectives

Atrial fibrillation (AF)–associated ischemic stroke is often managed as a single clinical entity; however, prognosis may vary depending on whether AF was detected after the stroke or whether the stroke occurred despite prior oral anticoagulant (OAC) use. We aimed to describe long-term outcomes after ischemic stroke across 3 distinct AF phenotypes: AF complicated by ischemic stroke despite prior anticoagulation (AFIDA), AF complicated by ischemic stroke without prior anticoagulation (OAC-naive AF), and AF detected after stroke (AFDAS).

Methods

This nationwide cohort study used the DeSC-IQVIA database, an administrative claims database covering approximately 16 million individuals in Japan. Adult stroke survivors with AF discharged with OAC prescriptions between April 2014 and January 2025 were included. Three mutually exclusive AF phenotypes defined by AF detection timing and prior anticoagulation status: AFIDA, OAC-naive AF, and AFDAS (diagnosed during the index hospitalization). The primary outcomes were hospitalization for recurrent ischemic stroke or systemic embolism (SE). The secondary outcomes included major bleeding, heart failure (HF) hospitalization, and all-cause mortality. Cumulative incidence was estimated using the Aalen-Johansen method, accounting for the competing risk of death. Adjusted subdistribution hazard ratios (aSHRs) were estimated using Fine-Gray models.

Results

Among 21,586 patients (median age, 83 years [interquartile range (IQR), 78–88 years]; 10,604 [49.1%] female), 6,604 (30.6%) were classified as having AFIDA, 11,875 (55.0%) as having OAC-naive AF, and 3,107 (14.4%) as having AFDAS. During the 38,593 person-years of follow-up, 2,028 patients experienced stroke/SE. The 5-year cumulative incidence of stroke/SE was highest in AFIDA (18.6% [95% CI 17.0%–20.2%]), followed by OAC-naive AF (13.0% [95% CI 12.0%–13.9%]) and AFDAS (10.5% [95% CI 9.0%–12.1%]). Compared with OAC-naive AF, AFIDA was associated with a higher risk of recurrent stroke/SE (aSHR, 1.38; 95% CI 1.25–1.52), whereas AFDAS was associated with a lower risk (aSHR, 0.87; 95% CI 0.75–1.00). Among secondary outcomes, AFIDA showed an increased risk of HF hospitalization (aSHR, 1.15; 95% CI 1.02–1.31).

Discussion

Long-term prognosis after AF-associated stroke is heterogeneous across AF phenotypes, with AFIDA representing a high-risk group and AFDAS representing a low-risk group. These findings highlight the need to treat these clinically identifiable phenotypes as distinct target populations for secondary prevention strategies and future clinical trials.

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Glossary

AF = atrial fibrillation; **AFDAS** = AF detected after stroke; **AFIDA** = AF complicated by ischemic stroke despite prior oral anticoagulation; **aSHR** = adjusted subdistribution hazard ratio; **DPC** = diagnosis procedure combination; **HF** = heart failure; **ICD-10** = International Classification of Diseases, Tenth Revision; **ICM** = insertable cardiac monitor; **IQR** = interquartile range; **OAC** = oral anticoagulant; **OAC-naive AF** = AF complicated by ischemic stroke without prior anticoagulation; **SE** = systemic embolism; **SMD** = standardized mean differences.

Introduction

Atrial fibrillation (AF) is a major cause of ischemic stroke, accounting for 20%–30% of all cases.[1,2] Although the widespread use of oral anticoagulants (OACs) has standardized stroke prevention, ischemic stroke cannot be fully prevented, and the residual risk remains a clinical challenge. Previous studies have suggested that approximately 1 in 6 patients experience recurrence within 5 years.[3] However, this risk is heterogeneous, highlighting the limitations of treating AF-associated stroke as a single condition.

Clinically, 3 distinct phenotypes are recognized based on the timing of AF detection and prior anticoagulation status: (1) AF complicated by ischemic stroke despite prior oral anticoagulation (AFIDA), (2) AF complicated by ischemic stroke without prior oral anticoagulation (OAC-naive AF), and (3) AF detected after stroke (AFDAS).[4,5] AFIDA represents a high-risk phenotype characterized by a suboptimal response to anticoagulation, with a 1.5-fold higher annual recurrence risk than OAC-naive AF.[6] Conversely, AFDAS has been estimated to carry a 21%–26% lower risk of recurrence than previously known AF; however, whether AF was causal, incidental, or triggered by the stroke itself is often unclear.[7,8] Despite their growing clinical relevance, robust epidemiologic data directly comparing the long-term outcomes among these groups within a single cohort are lacking, hindering the evidence-based planning of clinical trials targeting these specific subgroups. This study aimed to describe the long-term outcomes of patients with ischemic stroke with AF across the 3 phenotypes.

Methods

Study Design

We conducted a nationwide cohort study using routinely collected administrative claims data to describe long-term outcomes after ischemic stroke with AF across the 3 phenotypes. To ensure comparability across phenotypes and to focus on the secondary prevention phase, we aligned a common time zero at hospital discharge from the index stroke hospitalization and evaluated outcomes occurring after discharge. This study followed the Reporting of Studies Conducted Using Observational Routinely Collected Health Data statement.[9]

Data Source

We used the DeSC-IQVIA integrated database (DeSC Healthcare, Tokyo, Japan), a commercial administrative claims database covering April 1, 2014, to January 31, 2025, including approximately 16 million individuals.[10] The database consolidates claims from 3 major Japanese insurance schemes: the corporate health insurance (employees and their dependents aged <75 years), the national health insurance (primarily self-employed, unemployed, or retired individuals aged <75 years), and the late-stage medical care system for older adults (aged ≥75 years). This database is designed to reflect the demographic structure of the Japanese population, and the prevalence of major comorbidities was similar to that in a previously reported national survey.[11] The database contains anonymized demographics, diagnosis codes (International Classification of Diseases, Tenth Revision [ICD-10]), procedures, and dispensed prescriptions, enabling longitudinal follow-up while individuals remain enrolled with the same insurer (details in eMethods 1).

This study's key feature was the use of inpatient claims linked to the diagnosis procedure combination (DPC) system, a nationwide case-mix classification system covering approximately 90% of the tertiary facilities in Japan.[12,13] The DPC system includes diagnosis-timing indicators that distinguish conditions present at admission from those diagnosed after admission. We leveraged this feature to classify the timing of AF diagnosis relative to the index stroke, minimizing misclassification inherent in conventional claims data. DPC linkage was required only for the index hospitalization to determine diagnosis timing; follow-up outcomes were captured using all inpatient claims, regardless of DPC status.

Standard Protocol Approvals, Registrations, and Patient Consents

This study protocol was approved by the Ethics Committee of Kyoto University Graduate School and Faculty of Medicine; informed consent was waived because only anonymized data were used (Approval No. R5455).

Patient Selection

We identified patients aged ≥18 years hospitalized for acute ischemic stroke during the study period. Index hospitalization was defined as the first admission with a primary diagnosis of ischemic stroke, accompanied by acute stroke care procedures to improve specificity. Patients were eligible if they met all of the following criteria: (1) valid diagnosis-timing indicators available through DPC-linked claims; (2) coverage by an insurer with an available mortality flag; (3) continuous insurance enrollment for at least 365 days before admission to establish baseline characteristics and exclude recent stroke; and (4) plausible follow-up duration (≥0 days) to avoid data recording errors. Patients with diagnoses suggesting alternative stroke mechanisms and those who underwent percutaneous/surgical left atrial appendage occlusion (LAAO) before the index stroke were excluded. From this cohort, we identified patients with AF recorded during the index hospitalization, then excluded those who died during hospitalization or were discharged without an OAC prescription. Although these final criteria excluded patients who died during index hospitalization or transitioned to palliative care without anticoagulation, this restriction aligns with this study's objective of evaluating secondary prevention strategies in stroke survivors. Detailed operational definitions are provided in eTable 1.

AF Phenotype Definition

Patients were classified into 3 mutually exclusive phenotypes based on diagnosis-timing indicators and preadmission OAC status.

1. *AFIDA:* AF was documented as a comorbid condition at admission, with evidence of OAC prescription coverage on the admission date.
2. *OAC-naive AF:* AF documented as a comorbid condition at admission without evidence of OAC prescription coverage on the admission date.
3. *AFDAS:* AF was recorded as a postadmission diagnosis during index hospitalization.

OAC prescription coverage on the admission date was determined using dispensing records from the preceding 180 days; prescriptions dispensed on the admission date were excluded as these may reflect newly initiated therapy. The AFDAS definition was restricted to AF recorded during the index hospitalization to align the time zero across phenotypes. Another important subset of AFDAS, AF detected after discharge, often through an insertable cardiac monitor (ICM), was not captured by the definition and may represent a distinct and potentially lower-risk phenotype.[14] Therefore, "ICM-detected AFDAS" was evaluated separately as an exploratory analysis (eMethods 2, eTable 2).

Outcomes

The primary outcome was hospitalization for ischemic stroke or systemic embolism (SE). The secondary outcomes included hospitalization for major bleeding, hospitalization for heart failure (HF), and all-cause mortality. Major bleeding events included intracranial hemorrhage, gastrointestinal bleeding, and other major extracranial bleeding events. All hospitalization outcomes required primary diagnosis coding to enhance specificity. Detailed outcome definitions and code lists are provided in eTable 3.

Follow-Up and Time Zero

Time zero was defined as discharge from the index stroke hospitalization, aligning the start of observation across all phenotypes. Patients were followed up from time zero until the first occurrence of an outcome event, insurance disenrollment, death, or administrative censoring (January 31, 2025), whichever occurred first.

Covariates

We identified potential confounders based on prior studies.[15] Baseline comorbidities were ascertained from the ICD-10 diagnostic codes recorded from 365 days to 1 day before the discharge date. Comorbidities of interest were defined as the components of the CHA_2DS_2-VASc and the HAS-BLED score, including other relevant conditions (eTable 4). Baseline medication use was identified from outpatient and inpatient dispensing records from 180 days to 1 day before the discharge date. The medications of interest included antiplatelet agents (aspirin and P2Y12 inhibitors), rhythm control agents (Vaughan Williams class Ia, Ic, and III), beta-blockers, calcium channel blockers, renin-angiotensin-aldosterone system inhibitors, proton pump inhibitors, corticosteroids, nonsteroidal anti-inflammatory drugs, diuretics, insulin or oral antidiabetic drugs, and thyroid replacements. The type of OAC prescribed at discharge (direct oral anticoagulants vs warfarin) was also recorded as a covariate (eTable 5).

Statistical Analysis

Baseline characteristics were compared among the 3 phenotypes using standardized mean differences (SMD), with an absolute SMD >0.1 considered to indicate imbalance.[16] The incidence rates for all outcomes were calculated as events per 100 person-years with 95% CIs. To describe absolute risks, this study estimated the cumulative incidence probabilities at 1 and 5 years as primary time points, with annual estimates over 5 years. The Aalen-Johansen estimator for nonfatal outcomes (recurrent thromboembolism, major bleeding, and HF hospitalization) was used to account for the competing risk of death, and the Kaplan-Meier estimator for all-cause mortality.[17] Between-group differences were assessed using Gray's test and log-rank test, respectively.

Between-phenotype comparisons were considered supplementary analyses to contextualize the prognostic profiles of AFIDA and AFDAS relative to OAC-naive AF. To estimate the adjusted associations over the entire follow-up period, multivariable regression models were fitted using OAC-naive AF as the reference. For nonfatal outcomes, Fine-Gray subdistribution hazard models were used to estimate adjusted subdistribution hazard ratios (aSHRs), treating death as a competing event. For all-cause mortality, Cox proportional hazards models were used to estimate the adjusted hazard ratios (aHRs). All models were adjusted for the full set of prespecified baseline covariates. Baseline comorbidities and medications were defined based on the presence of relevant codes during the lookback period; therefore, the absence of codes was treated as the absence of the condition. Other covariates were complete in the included population, and multiple imputations were not performed.

All analyses were performed using Stata version 16.0 (StataCorp). We interpreted results based on point estimates and their uncertainty rather than *p*-value thresholds. Several sensitivity analyses were conducted to address database-specific characteristics and to examine alternative outcome definitions (eMethods 1). To characterize broader AF management trajectories, we also conducted exploratory analyses of ICM-detected AFDAS and percutaneous LAAOs (eMethods 2).

Data Availability

Data are not available for sharing due to licensing restrictions of the commercial database (DeSC Healthcare, Tokyo, Japan). Interested researchers may contact the database provider directly for data access.

Results

Study Population and Patient Characteristics

In total, 21,586 eligible patients were identified for the main analysis (Figure 1). The median age was 83 years (interquartile range [IQR], 78–88 years), and 10,604 patients (49.1%) were female. Among these, 6,604 (30.6%) were classified as having AFIDA, 11,875 (55.0%) as having OAC-naive AF, and 3,107 (14.4%) as having AFDAS. The total person-time was 38,593 person-years (11,733 person-years for AFIDA, 21,182 person-years for OAC-naive AF, and 5,678 person-years for AFDAS). The median follow-up was 1.44 years (IQR, 0.60–2.72) overall: 1.46 years (IQR, 0.59–2.71) for AFIDA, 1.42 years (IQR, 0.59–2.72) for OAC-naive AF, and 1.48 years (IQR, 0.63–2.75) for AFDAS. The baseline characteristics of the 3 phenotypes are presented in Table 1. The mean age was similar across groups (82.1–82.6 years). The CHA_2DS_2-VASc score was highest in AFIDA (6.0), followed by AFDAS (5.8) and OAC-naive AF (5.7). HF was more prevalent in patients with AFIDA (50.2%) than in those with OAC-naive AF (39.5%) or AFDAS (36.0%). Beta-blockers were more commonly prescribed for AFIDA (57.4%) than for OAC-naive AF (44.5%) or AFDAS (43.4%). Conversely, antiplatelet agents were more frequently used in AFDAS (48.2%) than in AFIDA (31.8%) or OAC-naive AF (33.4%), consistent with the clinical scenario in which AF is not yet recognized before stroke.

Primary Outcome

During follow-up, 2,028 patients experienced the primary outcome of hospitalization for ischemic stroke or SE (776 events in the AFIDA group, 1,015 in the OAC-naive AF group, and 237 in the AFDAS group). Most of the events were recurrent ischemic stroke (1,921 [94.7%]), with SE accounting for 107 events (5.3%). Incidence rates differed across phenotypes: 7.28 per 100 person-years (95% CI 6.78–7.81) in AFIDA, 5.13 (95% CI 4.82–5.45) in OAC-naive AF, and 4.43 (95% CI 3.88–5.03) in AFDAS. The cumulative incidence was the highest in the AFIDA group at all time points (Figure 2A). At 1 year, cumulative incidence was 8.3% (95% CI 7.6%–9.1%) in AFIDA, 6.3% (95% CI 5.8%–6.8%) in OAC-naive AF, and 5.7% (95% CI 4.9%–6.6%) in AFDAS. At 5 years, cumulative incidence reached 18.6% (95% CI 17.0%–20.2%), 13.0% (95% CI 12.0%–13.9%), and 10.5% (95% CI 9.0%–12.1%), respectively (Table 2). In multivariable Fine-Gray models with OAC-naive AF as the reference, AFIDA was associated with a higher risk of recurrent thromboembolism (aSHR, 1.38; 95% CI 1.25–1.52), whereas AFDAS was associated with a lower risk (aSHR, 0.87; 95% CI 0.75–1.00) (Table 3). Sensitivity analyses showed consistent results (eTables 6 and 7). The results of the exploratory analyses are presented in eTables 8, 9, and eFigures 1–4.

Secondary Outcomes

Major Bleeding

In total, 494 patients (168 with AFIDA, 259 with OAC-naive AF, and 67 with AFDAS) were hospitalized for major bleeding. The point estimate of the incidence rate was higher in AFIDA (1.46 per 100 person-years; 95% CI 1.25–1.70) than in OAC-naive AF (1.24; 95% CI 1.09–1.40) or AFDAS (1.20; 95% CI 0.93–1.52). The 1-year cumulative incidence ranged from 1.4% to 1.7%, and the 5-year cumulative incidence ranged from 3.4% to 4.2% across the phenotypes (Table 2; Figure 2B). In multivariable Fine-Gray models for major bleeding, neither AFIDA (aSHR, 1.16; 95% CI 0.95–1.42) nor AFDAS (aSHR, 0.95; 95% CI 0.73–1.24) differed from OAC-naive AF (Table 3).

HF

Hospitalization for HF occurred in 1,209 patients (445 with AFIDA, 620 with OAC-naive AF, and 144 with AFDAS). AFIDA had the highest incidence rate (3.96 per 100 person-years; 95% CI 3.60–4.34), followed by OAC-naive AF (3.03; 95% CI 2.80–3.28) and AFDAS (2.62; 95% CI 2.21–3.09). At 1 year, the cumulative incidence was 4.2% in AFIDA, 3.4% in OAC-naive AF, and 2.8% in AFDAS; at 5 years, these rates were 11.1%, 8.6%, and 7.4%, respectively (Table 2; Figure 2C). In multivariable Fine-Gray models for HF hospitalization, AFIDA showed increased risk (aSHR, 1.15; 95% CI 1.02–1.31), while AFDAS did not differ from OAC-naive AF (aSHR, 0.90; 95% CI 0.75–1.09) (Table 3).

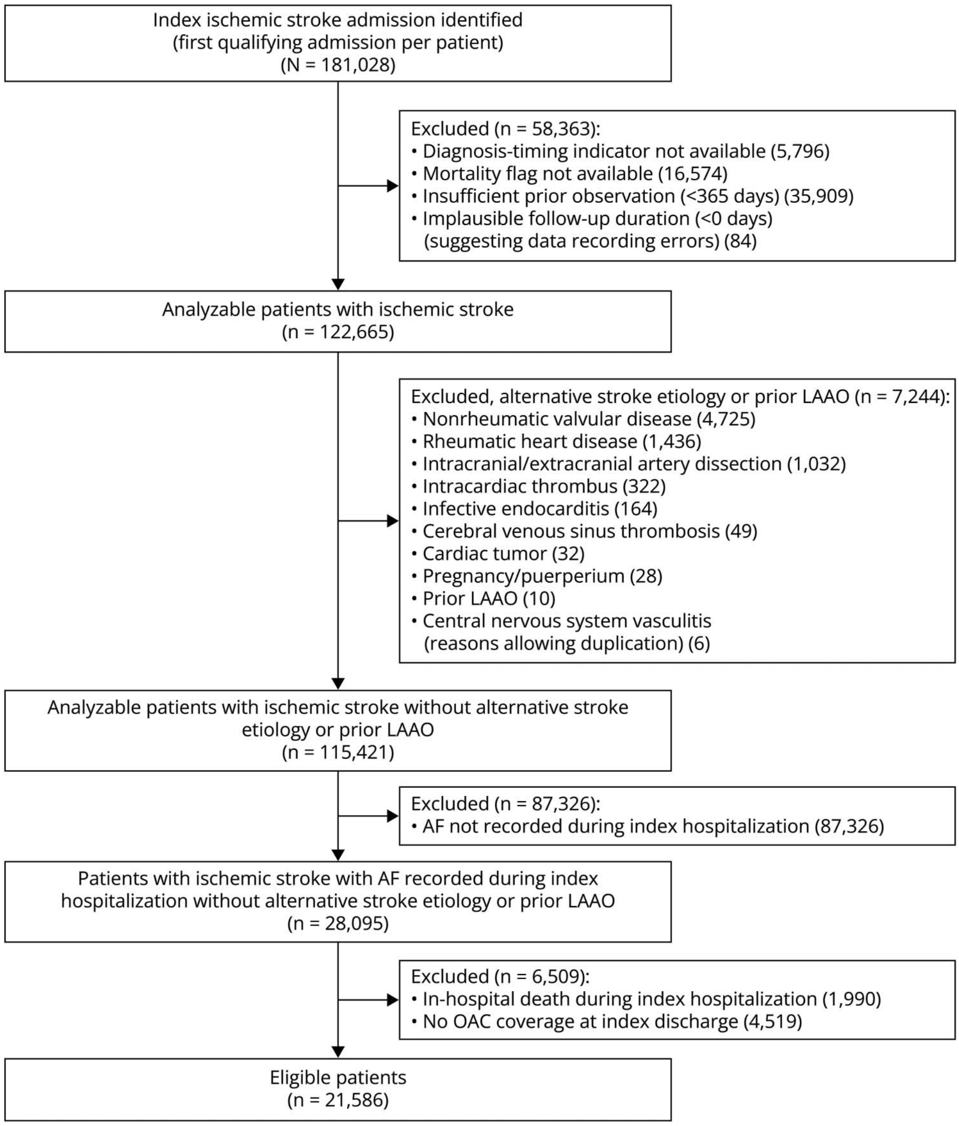
All-Cause Mortality

During follow-up, 6,536 patients (30.3%) died (1,973 with AFIDA, 3,651 with OAC-naive AF, and 912 with AFDAS). Incidence rates were similar across phenotypes: 16.8 per 100 person-years (95% CI 16.1–17.6) in AFIDA, 17.2 (95% CI 16.7–17.8) in OAC-naive AF, and 16.1 (95% CI 15.0–17.1) in AFDAS. At 1 year, Kaplan-Meier estimates ranged from 16.6% to 18.2%; at 5 years, approximately half of the patients in each group died (52.5%–55.1%) (Table 2; Figure 2D). For all-cause mortality, AFIDA showed a lower risk after adjustment (aHR, 0.90; 95% CI 0.85–0.95), whereas AFDAS did not differ from OAC-naive AF (aHR, 0.99; 95% CI 0.92–1.07) (Table 3).

Discussion

In this nationwide cohort study in Japan, we described long-term outcomes after ischemic stroke with AF across 3 phenotypes based on detection timing and prior anticoagulation status.

Figure 1 Study Flowchart



AF = atrial fibrillation; LAAO = left atrial appendage occlusion; OAC = oral anticoagulant.

Compared with OAC-naive AF, AFIDA was associated with a higher risk of recurrent thromboembolism (aSHR, 1.38; 95% CI 1.25–1.52), whereas AFDAS was associated with a lower risk (aSHR, 0.87; 95% CI 0.75–1.00). Five-year cumulative incidences of AFIDA, OAC-naive AF, and AFDAS were 18.6%, 13.0%, and 10.5%, respectively. The risk of heart failure hospitalization was also elevated in patients with AFIDA, although adjusted all-cause mortality was paradoxically lower. These findings highlight the need to treat these clinically identifiable phenotypes as distinct target populations for secondary prevention strategies and future clinical trials.

AFIDA showed a higher recurrence risk than OAC-naive AF, consistent with previous studies.[4,18] The aSHR of 1.38 in our study was more modest than the HR of 1.6 (95% CI 1.2–2.3) from the pooled analysis of 7 international cohorts.[4] This attenuation likely reflects differences in comparator composition; previous studies combined OAC-naive patients with lower-risk AFDAS in the comparator group and adjusted for AF detection timing as a covariate, but such adjustment may not fully account for the distinct risk profile of AFDAS.[4] The mechanisms underlying AFIDA are likely multifactorial.[19] One proposed mechanism is advanced atrial cardiopathy; the higher prevalence of HF in the AFIDA group (50.2% vs 39.5% in OAC-naive AF) may reflect more advanced atrial substrate disease.[20,21] Other proposed mechanisms include subtherapeutic anticoagulation intensity, true pharmacologic resistance, and coexisting noncardioembolic mechanisms such as large artery atherosclerosis.[19] Although claims data preclude a direct assessment of these mechanisms, our results remained robust after excluding patients with intra/extracranial artery stenosis.

AFDAS demonstrated a lower recurrence rate than OAC-naive AF, which is consistent with the results of prior studies.

Table 1 Baseline Characteristics by Atrial Fibrillation Phenotype

Characteristic	AFIDA (n = 6,604)	OAC-naive AF (n = 11,875)	AFDAS (n = 3,107)	SMD AFIDA vs OAC-naive AF	SMD OAC-naive AF vs AFDAS	SMD AFDAS vs AFIDA
Demographics						
Age, mean (SD), y	82.3 (6.6)	82.6 (8.0)	82.1 (7.9)	−0.04	0.06	−0.03
Female sex	2,851 (43.2)	6,149 (51.8)	1,604 (51.6)	−0.17	0.00	0.17
Risk scores						
CHA_2DS_2-VASc score, mean (SD)	6.0 (1.12)	5.7 (1.16)	5.8 (1.17)	0.18	−0.04	−0.14
Modified HAS-BLED score, mean (SD)[a]	3.8 (0.97)	3.6 (0.95)	3.8 (0.90)	0.15	−0.17	0.02
Baseline comorbidities						
Heart failure	3,313 (50.2)	4,686 (39.5)	1,119 (36.0)	0.22	0.07	−0.29
Hypertension	4,911 (74.4)	8,362 (70.4)	2,341 (75.4)	0.09	−0.11	0.02
Prior stroke/TIA/SE	6,604 (100)	11,875 (100)	3,107 (100)	—	—	—
Diabetes mellitus	2,026 (30.7)	2,711 (22.8)	815 (26.2)	0.18	−0.08	−0.10
Vascular disease	531 (8.0)	784 (6.6)	207 (6.7)	0.06	0.00	−0.05
Abnormal kidney function	802 (12.1)	1,029 (8.7)	239 (7.7)	0.11	0.04	−0.15
Abnormal liver function	463 (7.0)	785 (6.6)	197 (6.3)	0.02	0.01	−0.03
Bleeding predisposition	1,613 (24.4)	2,262 (19.0)	633 (20.4)	0.13	−0.03	−0.10
Alcohol abuse	44 (0.7)	78 (0.7)	13 (0.4)	0.00	0.03	−0.03
Carotid/intracranial stenosis	1,237 (18.7)	2,511 (21.1)	717 (23.1)	−0.06	−0.05	0.11
Cardiomyopathy	99 (1.5)	93 (0.8)	27 (0.9)	0.07	−0.01	−0.06
Anemia	1,147 (17.4)	1,640 (13.8)	464 (14.9)	0.10	−0.03	−0.07
Cancer (nonmetastatic)	720 (10.9)	1,041 (8.8)	319 (10.3)	0.07	−0.05	−0.02
Cancer, metastatic	79 (1.2)	87 (0.7)	40 (1.3)	0.05	−0.06	0.01
Dyslipidemia	2,900 (43.9)	4,638 (39.1)	1,522 (49.0)	0.10	−0.20	0.10
Fracture	1,504 (22.8)	2,469 (20.8)	650 (20.9)	0.05	0.00	−0.05
Peptic ulcer/duodenitis/gastritis	2,067 (31.3)	3,571 (30.1)	985 (31.7)	0.03	−0.04	0.01
Pulmonary disease	1,450 (22.0)	2,524 (21.3)	672 (21.6)	0.02	−0.01	−0.01
Baseline medications						
Antiplatelet agents	2,097 (31.8)	3,961 (33.4)	1,497 (48.2)	−0.03	−0.31	0.34
Rhythm control agents (class Ia/Ic/III)	334 (5.1)	437 (3.7)	106 (3.4)	0.07	0.02	−0.08
Beta-blockers	3,788 (57.4)	5,282 (44.5)	1,347 (43.4)	0.26	0.02	−0.28
Calcium channel blockers	4,240 (64.2)	7,663 (64.5)	2,064 (66.4)	−0.01	−0.04	0.05
RAAS inhibitors	3,511 (53.2)	5,414 (45.6)	1,552 (50.0)	0.15	−0.09	−0.06
Proton pump inhibitors	4,965 (75.2)	9,009 (75.9)	2,483 (79.9)	−0.02	−0.10	0.11
Corticosteroids	833 (12.6)	1,365 (11.5)	438 (14.1)	0.03	−0.08	0.04
NSAIDs	2,485 (37.6)	4,367 (36.8)	1,233 (39.7)	0.02	−0.06	0.04
Insulin or oral antidiabetic drugs	1,912 (29.0)	2,497 (21.0)	767 (24.7)	0.18	−0.09	−0.10

Continued

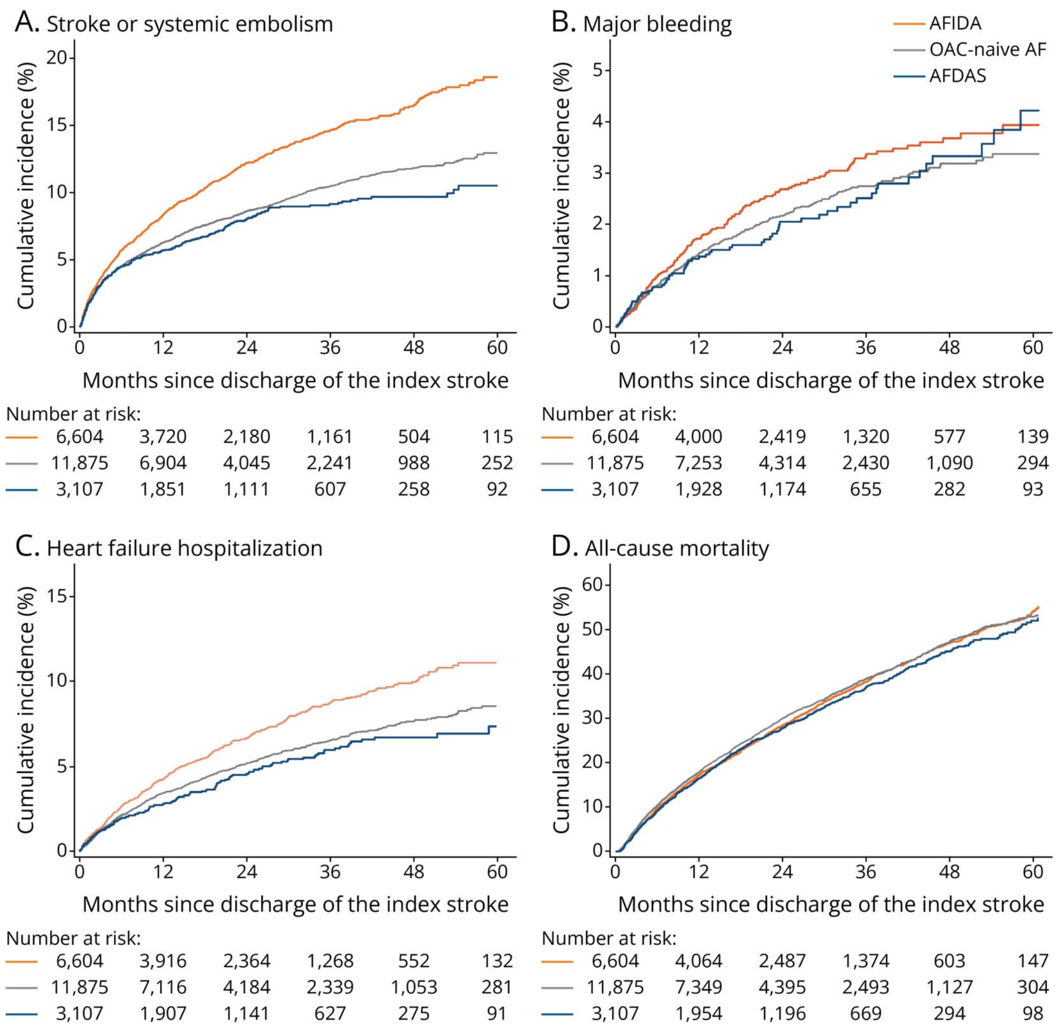
Table 1 Baseline Characteristics by Atrial Fibrillation Phenotype *(continued)*

Characteristic	AFIDA (n = 6,604)	OAC-naive AF (n = 11,875)	AFDAS (n = 3,107)	SMD AFIDA vs OAC-naive AF	SMD OAC-naive AF vs AFDAS	SMD AFDAS vs AFIDA
Thyroid replacement	62 (0.9)	62 (0.5)	11 (0.4)	0.05	0.03	−0.07
OAC type at discharge: DOAC (vs warfarin)	5,894 (89.3)	11,261 (94.8)	2,993 (96.3)	−0.21	−0.07	0.28

Abbreviations: AF = atrial fibrillation; AFDAS = AF detected after stroke; AFIDA = AF complicated by ischemic stroke despite prior anticoagulation; DOAC = direct oral anticoagulant; NSAIDs = nonsteroidal anti-inflammatory drugs; OAC = oral anticoagulant; OAC-naive AF = AF complicated by ischemic stroke without prior anticoagulation; RAAS = renin–angiotensin–aldosterone system; SE = systemic embolism; SMD = standardized mean difference.
SMD columns: absolute SMD >0.1 indicates a significant imbalance.
Data are presented as No. (%) unless otherwise indicated.
[a] As the labile international normalized ratio could not be ascertained from the claims data, we calculated the HAS-BLED score using available components.

The aSHR of 0.87 in our study was more modest than the 21%–26% relative risk reductions previously reported.[7,8] Two methodological factors may explain this difference. First, many previous AFDAS studies either set time zero at AF detection or did not explicitly specify the follow-up start point.[22-24] When time zero is delayed to AF detection only in

Figure 2 Cumulative Incidence of Clinical Outcomes by Atrial Fibrillation Phenotype



Cumulative incidence curves for clinical outcomes stratified by atrial fibrillation (AF) phenotype: AF complicated by ischemic stroke despite prior anticoagulation (AFIDA; orange), AF complicated by ischemic stroke without prior anticoagulation (OAC-naive AF; gray), and AF detected after stroke (AFDAS; blue). Panel A shows the cumulative incidence of stroke or systemic embolism estimated using the Aalen-Johansen method, accounting for the competing risk of death. Panel B shows the cumulative incidence of major bleeding events. Panel C shows the cumulative incidence of hospitalization because of heart failure. Panel D shows the all-cause mortality estimated using the Kaplan-Meier method. AF = atrial fibrillation; AFDAS = AF detected after stroke; AFIDA = AF complicated by ischemic stroke despite prior anticoagulation; OAC = oral anticoagulant.

Table 2 Incidence Rates and Cumulative Incidence of Clinical Outcomes

Outcome	Incidence rate/100 PY (95% CI)	1-Year % (95% CI)	2-Year % (95% CI)	3-Year % (95% CI)	4-Year % (95% CI)	5-Year % (95% CI)
Stroke/SE						
AFIDA	7.28 (6.78–7.81)	8.3 (7.6–9.1)	12.2 (11.3–13.1)	14.7 (13.6–15.7)	16.5 (15.3–17.7)	18.6 (17.0–20.2)
OAC-naive AF	5.13 (4.82–5.45)	6.3 (5.8–6.8)	8.6 (8.1–9.2)	10.5 (9.8–11.1)	11.8 (11.1–12.6)	13.0 (12.0–13.9)
AFDAS	4.43 (3.88–5.03)	5.7 (4.9–6.6)	8.1 (7.0–9.1)	9.2 (8.0–10.3)	9.7 (8.4–11.0)	10.5 (9.0–12.1)
Major bleeding						
AFIDA	1.46 (1.25–1.70)	1.7 (1.4–2.1)	2.7 (2.2–3.1)	3.4 (2.8–3.9)	3.7 (3.1–4.3)	3.9 (3.2–4.6)
OAC-naive AF	1.24 (1.09–1.40)	1.5 (1.2–1.7)	2.2 (1.9–2.5)	2.7 (2.4–3.1)	3.2 (2.8–3.6)	3.4 (2.9–3.9)
AFDAS	1.20 (0.93–1.52)	1.4 (0.9–1.8)	2.0 (1.5–2.6)	2.5 (1.8–3.2)	3.3 (2.4–4.2)	4.2 (2.9–5.6)
Heart failure						
AFIDA	3.96 (3.60–4.34)	4.2 (3.7–4.7)	6.7 (6.0–7.3)	8.7 (7.9–9.6)	10.0 (9.0–11.0)	11.1 (9.9–12.3)
OAC-naive AF	3.03 (2.80–3.28)	3.4 (3.1–3.8)	5.2 (4.7–5.6)	6.5 (6.0–7.1)	7.6 (7.0–8.3)	8.6 (7.7–9.4)
AFDAS	2.62 (2.21–3.09)	2.8 (2.2–3.4)	4.6 (3.7–5.4)	6.0 (4.9–7.0)	6.7 (5.5–7.9)	7.4 (5.9–8.9)
All-cause mortality						
AFIDA	16.82 (16.08–17.57)	17.4 (16.5–18.4)	28.6 (27.4–30.0)	38.6 (37.1–40.2)	47.3 (45.4–49.2)	55.1 (52.5–57.8)
OAC-naive AF	17.24 (16.68–17.80)	18.2 (17.4–19.0)	30.2 (29.3–31.2)	39.3 (38.1–40.4)	47.8 (46.5–49.2)	53.2 (51.5–55.0)
AFDAS	16.06 (15.04–17.14)	16.6 (15.2–18.0)	28.0 (26.2–29.9)	37.4 (35.3–39.7)	45.8 (43.1–48.6)	52.5 (48.9–56.1)

Abbreviations: AF = atrial fibrillation; AFDAS = AF detected after stroke; AFIDA = AF complicated by ischemic stroke despite prior anticoagulation; OAC-naive AF = AF complicated by ischemic stroke without prior anticoagulation; PY = person-years; SE = systemic embolism.
Cumulative incidence was estimated using the Aalen-Johansen estimator, which accounts for the competing risk of death. The Kaplan-Meier estimates are shown for all-cause mortality.

the AFDAS group, an immortal time bias arises in favor of AFDAS; patients must, by definition, remain event-free until AF is detected. Second, prior studies commonly defined the control group as having "known AF," which typically included both OAC-naive patients and high-risk AFIDA patients.[23] Contamination by AFIDA increases the event rate in the control group, thereby amplifying the apparent low risk associated with AFDAS.

Several hypotheses have been proposed to explain these favorable outcomes. First, the initial stroke in some AFDAS cases may not have been caused by AF. In these patients, AF detected after stroke may reflect an evolving atrial cardiopathy that had not yet manifested as detectable AF at the time of stroke, or the stroke may have been caused by a different mechanism altogether. Second, the AF burden thresholds may be relevant. The ASSERT trial showed that only subclinical AF episodes exceeding 24 hours significantly increased stroke risk, whereas shorter episodes did not.[25] Given that 52% of AFDAS episodes last <30 seconds, such brief episodes may not confer a stroke risk equivalent to sustained AF.[26] These factors may partially explain the lower AFDAS recurrence rate.

In this study, hospitalization for HF and thromboembolic recurrence varied in parallel across phenotypes; both outcomes were highest in the AFIDA group and lowest in the AFDAS group. HF in patients with AF is driven by an excessive AF burden and progressive atrial cardiopathy.[27] The parallel pattern observed in this study raises the possibility that thromboembolic recurrence is also driven by these factors. If so, rhythm control that attenuates the AF burden might reduce not only HF but also stroke/SE in this population. The ongoing EAST-STROKE trial is expected to provide important information regarding this question (NCT05293080).

The AFIDA group showed higher morbidity but lower adjusted mortality (HR 0.90; 95% CI 0.85–0.95). This paradoxical finding may reflect both clinical and methodological factors.

From a clinical perspective, prior anticoagulation has been associated with reduced stroke severity at onset, which may translate into better survival.[28] In addition, a healthy user effect may contribute. Maintaining long-term anticoagulation therapy before an index stroke requires sufficient cognitive function, medication adherence, and regular healthcare access. The AFIDA group may inherently select patients with these favorable characteristics.[29]

From a methodological perspective, several biases warrant consideration. First, all-cause mortality is a nonspecific

Table 3 Hazard Ratios for Clinical Outcomes by Atrial Fibrillation Phenotype

Outcome	Unadjusted SHR/HR (95% CI)	Adjusted SHR/HR (95% CI)
Stroke/SE		
OAC-naive AF	Reference	Reference
AFIDA	1.41 (1.28–1.55)	1.38 (1.25–1.52)
AFDAS	0.89 (0.77–1.02)	0.87 (0.75–1.00)
Major bleeding		
OAC-naive AF	Reference	Reference
AFIDA	1.18 (0.97–1.43)	1.16 (0.95–1.42)
AFDAS	0.98 (0.75–1.29)	0.95 (0.73–1.24)
Heart failure		
OAC-naive AF	Reference	Reference
AFIDA	1.31 (1.16–1.48)	1.15 (1.02–1.31)
AFDAS	0.88 (0.74–1.06)	0.90 (0.75–1.09)
All-cause mortality		
OAC-naive AF	Reference	Reference
AFIDA	0.97 (0.92–1.03)	0.90 (0.85–0.95)
AFDAS	0.93 (0.87–1.00)	0.99 (0.92–1.07)

Abbreviations: AF = atrial fibrillation; AFDAS = AF detected after stroke; AFIDA = AF complicated by ischemic stroke despite prior anticoagulation; HR = hazard ratio; OAC-naive AF = AF complicated by ischemic stroke without prior anticoagulation; SE = systemic embolism; SHR = subdistribution hazard ratio. Subdistribution hazard ratios from the Fine-Gray competing-risks regression are shown for stroke/SE, major bleeding, and heart failure, treating death as a competing event. Hazard ratios from Cox proportional hazards regression analysis are shown for all-cause mortality. Adjusted models included age, sex, comorbidities (heart failure, hypertension, diabetes, vascular disease, abnormal kidney function, abnormal liver function, bleeding predisposition, alcohol abuse, carotid/intracranial stenosis, cardiomyopathy, anemia, nonmetastatic cancer, metastatic cancer, dyslipidemia, fracture, peptic ulcer/duodenitis/gastritis, pulmonary disease), and baseline medications (antiplatelet agents, rhythm control agents, beta-blockers, calcium channel blockers, RAAS inhibitors, proton pump inhibitors, corticosteroids, NSAIDs, insulin or oral antidiabetic drugs, thyroid replacement, and oral anticoagulant type at discharge).

outcome, and our covariate adjustment was optimized for the primary outcome of stroke/SE rather than mortality; this may have introduced bias in the mortality estimates. Second, by focusing on postdischarge outcomes and requiring OAC prescription at discharge, this study excluded patients who died during the index hospitalization or transitioned to palliative care. Because AFIDA represents an advanced disease stage, these exclusion criteria may have disproportionately eliminated the most severe cases within that phenotype. Third, in this elderly cohort with approximately 50% 5-year all-cause mortality, high background noncardiovascular mortality likely diluted any mortality signal attributable to recurrent stroke or HF. Mortality was not a primary endpoint, and similar crude mortality across phenotypes did not undermine the main conclusion of phenotype-specific heterogeneity in thromboembolic recurrence risk.

This study has several limitations. First, the diagnosis-timing indicator cannot distinguish between known AF before hospitalization and AF first identified at the time of stroke presentation because both are classified as comorbidities at admission. However, data from Rizos et al.[30] suggested that the echocardiographic parameters did not differ substantially between these subgroups, indicating that this distinction may reflect an opportunity for detection rather than fundamental pathophysiologic differences. Second, important clinical information was unavailable, including AF burden, AF duration, AF type, echocardiographic parameters, stroke severity, anticoagulant dosing, medication adherence, and laboratory values such as renal function and natriuretic peptide levels. Furthermore, AF codes in administrative data likely capture symptomatic or longstanding AF rather than paroxysmal or subclinical AF, and the timing of AF diagnosis is a surrogate for the true timing of AF onset. Notably, up to 30% of new administrative AF diagnoses present with stroke as the first manifestation[31]; some of these patients may be included in our OAC-naive AF category. Nevertheless, such clinical information is often unavailable in resource-limited clinical settings or studies. Our phenotypic classification may serve as a pragmatic surrogate integrating these unmeasured factors. Third, our findings were subject to misclassification and under/overreporting. However, the internal validity was enhanced through specificity measures, such as procedure codes, strict inclusion criteria, and comprehensive sensitivity analyses, accounting for database limitations. Fourth, recurrent stroke was defined using all ischemic stroke codes regardless of etiology. Because administrative data cannot distinguish cardioembolic from other stroke subtypes, the AFIDA group may include recurrent strokes caused by non-AF mechanisms. This limitation may have partly contributed to the paradoxical finding that AFIDA showed higher stroke risk but lower mortality. Finally, the sample was restricted to the Japanese population; therefore, the generalizability to other countries/health systems is uncertain.

Long-term prognosis after AF-associated stroke is heterogeneous across phenotypes defined by the timing of AF detection and prior anticoagulation status, with AFIDA representing a high-risk group and AFDAS representing a low-risk group. These findings highlight the need to treat these clinically identifiable phenotypes as distinct target populations for secondary prevention strategies and future clinical trials.

Author Contributions

S. Egashira: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. I. Kosuke: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. M. Koga: drafting/revision of the manuscript for content, including medical writing for content. T. Toda: drafting/revision of the manuscript for content, including medical writing for content. Y. Imanaka: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design.

Study Funding

This study was supported by JSPS KAKENHI (JP23H00448) from the Japan Society for the Promotion of Science (Dr. Imanaka). The funders had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Disclosure

M. Koga reports honoraria from AstraZeneca, Bayer Yakuhin, Daiichi-Sankyo, Pfizer, BMS/Janssen Pharmaceuticals, and Otsuka Pharmaceutical, and research supports from Boston Scientific and Daiichi-Sankyo, all of which are outside of the submitted work. All other authors report no relevant disclosures. Go to Neurology.org/N for full disclosures.

Publication History

Received by *Neurology*® February 9, 2026. Accepted in final form April 1, 2026. Submitted and externally peer reviewed. The handling editor was Editor-in-Chief José Merino, MD, MPhil, FAAN.

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